

Accountability and Algorithmic Delegation: Experimental Evidence

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Abstract

Experimental evidence suggests that delegation can shift responsibility for adverse outcomes onto human intermediaries. We ask whether the same motive applies to delegation to algorithms. In a preregistered online experiment with 322 participants, a decision-maker chooses to make a payoff-relevant prediction for a matched recipient or delegate it to an algorithm, calibrated to perform equally well. We vary between subjects whether the recipient can impose punishment. The prospect of punishment reduces delegation by 18.3 percentage points, from 60.8% to 42.5%, reversing the direction that the responsibility-shifting account predicts. Delegation, in turn, offers no protection. Recipients condition punishment primarily on the realized outcome, not on whether the prediction was delegated. We discuss several candidate mechanisms for this reversal.

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1 Introduction

Algorithms shape a growing share of consequential decisions. Many of these decisions carry consequences that fall, at least in part, on others. Physicians rely on diagnostic systems, firms let screening software rank job applicants, and financial advisors draw on algorithmic tools. In each setting, a human decides whether to rely on the algorithm. Who bears responsibility when the resulting decision causes harm is the subject of a lively legal (Lemley and Casey, 2019; Buiten et al., 2023), ethical (Matthias, 2004; Santoni de Sio and Mecacci, 2021), and policy (European Commission, Directorate-General for Justice and Consumers, 2019; European Union, 2024) debate. Yet the normative question of where responsibility should lie is distinct from the behavioral questions of how responsibility is attributed and how those attributions shape behavior. Does delegating a decision to an algorithm change the responsibility that affected parties assign to the delegator? And does the prospect of being held responsible affect the decision to delegate? We provide experimental evidence on both questions.

For delegation between humans, the experimental literature points in a clear direction. Delegation shifts responsibility. Principals who delegate an unpopular decision to a human intermediary are punished less for it, feel less responsible for it, and delegate strategically for that reason (Bartling and Fischbacher, 2012; Hamman et al., 2010; Coffman, 2011; Oexl and Grossman, 2013). Recent evidence suggests that algorithms can inherit the intermediary’s role. Delegating to an algorithm helps decision-makers escape blame for adverse outcomes (Feier et al., 2022), lets them shed moral responsibility for self-serving choices (Tontrup and Sprigman, 2025) and makes dishonesty easier by obscuring intent (Köbis et al., 2025). If algorithms provide such cover, decision-makers who face potential punishment should be drawn to them, and delegation should protect the delegator when outcomes turn out badly. Yet no existing study varies whether delegators can be punished, leaving untested whether the prospect of punishment in fact motivates delegation.

We address both questions in a preregistered online experiment with 322 participants.¹ A decision-maker, Player A, first completes ten rounds of a prediction task for her own payoff, and then makes an additional prediction that determines the payoff of a matched recipient, Player B. Player A can either make the consequential prediction for Player B herself or delegate it to an algorithm calibrated to perform equally well at the task. Thus, the environment we study, a prediction task with outcome uncertainty performed by a decision-maker with no material stake in the affected party’s outcome, resembles the settings in which algorithmic delegation typically operates in practice. We then vary between subjects whether Player B can reduce Player A’s earnings. In the *Punishment* condition, we elicit Player B’s punishment for every combination of delegation choice and prediction outcome, providing a within-subject test of whether delegation insulates Player A from punishment, conditional on the outcome. In the *No-Punishment* condition, no such punishment is possible. With everything except the

¹The preregistration (AsPredicted #133980) is available at <https://aspredicted.org/hs9az8.pdf> and reproduced in full in Appendix C.

possibility of punishment held fixed, any difference in delegation rates is attributable to that possibility, allowing us to test directly whether the prospect of being held responsible moves the delegation decision.

We find that the prospect of punishment reverses the prediction based on the literature. When punishment is possible, 42.5 percent of decision-makers delegate to the algorithm. When it is not, 60.8 percent do. The possibility of punishment thus *reduces* delegation by 18.3 percentage points. Delegation, in turn, offers no protection against punishment. Conditional on the outcome, Player B punishes delegated and Player A’s own decisions similarly, and punishment responds almost entirely to the outcome of the prediction rather than to how it was made.

The possibility of punishment thus affects the decision to delegate, even though delegating draws no differential punishment. We examine several explanations to reconcile these findings. The first is anticipated punishment. Player A may keep the decision because she expects delegation to be punished more harshly. Her elicited beliefs provide no evidence of such an expectation and instead match how Player B actually punishes. The second is punishment-induced effort. Player A may keep the decision in hopes of outperforming the algorithm in the prediction task for Player B, in which case decision-makers should improve on their earlier performance precisely when punishment is possible. Instead, improvement concentrates where punishment is impossible. Third, by keeping the decision, Player A may aim to signal a willingness to try on Player B’s behalf, whereas delegation may appear as shirking. Because performance is noisy, the choice to retain control can carry this signal even without a measurable improvement in performance. However, the signal carries monetary value only if Player A expects apparent shirking to draw harsher punishment, which her beliefs do not indicate. Detached from actual punishment, Player A may instead care directly about Player B’s judgment. Because Player B can form such a judgment in both conditions, however, this concern can explain the treatment difference only if the prospect of punishment activates or strengthens it. Lastly, the data show selection in performance. It is disproportionately the better-performing Player As who retain the decision. Both an intrinsic preference for producing the good outcome oneself and a distaste for delegating uncertain decisions are consistent with this selection, but they operate in both conditions alike and cannot on their own generate the treatment difference. Thus, while the drop in delegation under punishment is robust, which motive produces it is a question we narrow down rather than settle.

Our contribution is twofold. First, we provide, to our knowledge, the first direct test of whether the prospect of punishment encourages delegation to an algorithm, as prior evidence on delegation to human intermediaries would predict (Bartling and Fischbacher, 2012; Hamman et al., 2010; Coffman, 2011; Oexl and Grossman, 2013). Our environment departs from these canonical designs in three respects. The intermediary is an algorithm rather than a person. The decision involves genuine uncertainty, so a bad outcome may reflect bad luck rather than bad intent. And its immediate consequences fall entirely on the recipient, so the underlying decision involves no tradeoff between the players’ payoffs and there is no self-serving choice to police. In

this environment, the effect runs in the opposite direction. The prospect of punishment discourages rather than encourages delegation, and it does so selectively among the decision-makers most likely to succeed. This reversal is the paper’s main empirical result, adding a determinant of preferences over algorithm use that operates through the decision context rather than through algorithm characteristics or the task.

Second, we use Player B’s punishment decision to examine how an affected party assigns responsibility for algorithmic delegation when the delegation choice is uninformative about relative ability. The closest precedent is [Feier et al. \(2022\)](#), whose treatments vary whether the intermediary is an algorithm or another participant under a fixed punishment regime, whereas we hold the intermediary fixed as an algorithm and vary whether punishment is possible. In their environment, decisions delegated to an algorithm are rewarded more generously than non-delegated decisions after bad outcomes. However, the same subject both makes a delegation choice and evaluates the decision, so that evaluations may partly reflect participants’ own delegation choices. Moreover, the delegation decision is strongly determined by beliefs about relative ability between oneself and the intermediary, inviting punishment of overconfidence rather than of delegation itself. Our design is intended to remove both channels by separating the delegator and evaluator roles and making delegation uninformative about relative ability. Once both channels are closed, the differential disappears, and punishment tracks the realized outcome instead.

The remainder of the paper is organized as follows. Section 2 relates the study to the literature. Section 3 presents the experimental design and hypotheses. Section 4 reports the results and examines candidate mechanisms. Section 5 concludes.

2 Related Literature

Delegation has long been studied as an efficiency-enhancing institutional arrangement in which principals rely on agents with better information, lower opportunity costs, or specialized skills.² Alongside these efficiency motives, the literature has identified a range of strategic motives. Delegation can serve, for instance, as a commitment device that alters the equilibrium ([Schelling, 1960](#); [Fershtman and Gneezy, 2001](#)), as a way to motivate the agent ([Aghion and Tirole, 1997](#)), or as a means of shifting responsibility for the outcome away from the principal, both in her own eyes (felt responsibility) and in the eyes of others (attributed blame).

Responsibility avoidance through delegation. The political-science and public-choice literatures recognized early that delegation can shift blame onto an intermediary ([Weaver, 1986](#); [Fiorina, 1986](#)). Experimental economics has established the same motive in incentivized settings, primarily using modified dictator games. [Bartling and Fischbacher \(2012\)](#) provide the canonical experimental evidence. In their design, a dictator may either choose between a fair and an unfair allocation of an endowment or instead delegate this choice to another person whose

²For an early overview of the principal–agent literature see, e.g., [Bolton and Dewatripont \(2005\)](#).

monetary incentives are aligned with the dictator's. A recipient, who is adversely affected by the unfair allocation, may then assign costly punishment to the dictator and the intermediary. They find that delegation effectively shifts punishment from the dictator to the intermediary, and the prospect of avoiding punishment constitutes a strong motive for the delegation decision itself. Complementary experiments suggest that the reduction in the principal's punishment stems from the absence of direct interaction with the affected party rather than from a pure transfer of blame to a deciding intermediary. The principal escapes punishment even when the intermediary remains completely passive, but not when the principal continues to interact with the recipient directly (Coffman, 2011). The blame the intermediary absorbs, by contrast, attaches to the act of choosing rather than to any contribution to the outcome. Punishment lands on the intermediary even when delegation itself has eliminated the fair option, leaving him only the choice between two unfair allocations (Oexl and Grossman, 2013). Consistent with blame attaching to the act of choosing, randomization devices provide the principal weaker cover than deciding humans (Bartling and Fischbacher, 2012; Oexl and Grossman, 2013) and attract correspondingly less delegation (Bartling and Fischbacher, 2012). The responsibility-shifting motive does not require a punishment stage. Hamman et al. (2010) find that principals' transfers to recipients fall sharply under delegation even without any punishment opportunity, and that principals report feeling less responsible for the resulting allocations, indicating that responsibility shifting also operates through felt responsibility and moral wiggle room.

Beyond the canonical dictator-game setting, the responsibility-shifting motive, and the strategic use of delegation it gives rise to, have been documented across a range of domains. Voters strategically avoid being pivotal, and non-pivotal voters are indeed punished less for the same group outcome (Bartling et al., 2015). Principals pay agents to lie on their behalf and deceive more readily through a delegate than directly (Erat, 2013; Kandul and Kirchkamp, 2018; Gawn and Innes, 2019). Observers rate delegating legislatures as less blameworthy, and do so even when the delegate is powerless to change the outcome (Hill, 2015). In financial decision-making, investors delegate financial decisions partly to shift blame (Holzmeister et al., 2023; Freer et al., 2026).

In sum, the literature documents that, in settings with self-interested choice, delegation reduces both the principal's punishment and her felt responsibility, in turn motivating delegation. Our setting departs from this canonical design along three dimensions. First, the intermediary is algorithmic rather than human. Second, the outcome falls entirely on a passive recipient rather than partly on the principal herself. Third, the underlying decision involves genuine predictive uncertainty rather than a known fair–unfair choice, removing the transparent self-serving intent that punishment sanctions. Whether the responsibility-shifting motive retains its force under these conditions is the question we take up.

Delegation to algorithms. A large empirical literature studies when and why people are willing to use algorithmic decision aids. Dietvorst et al. (2015) coin the term *algorithm aversion* for the finding that people prefer their own or other humans' judgment over algorithmic predic-

tions, even after observing that algorithms outperform human decision-makers.³ In contrast, Logg et al. (2019) document conditions under which subjects show *appreciation* of algorithmic advice over human advice. The picture that emerges is therefore one of substantial heterogeneity in algorithm preferences across tasks, contexts, and elicitation methods. Burton et al. (2020), Mahmud et al. (2022), Chugunova and Sele (2022), Qin et al. (2025) and Irlenbusch (2026) provide systematic reviews of this literature and document a large array of moderators for when and why algorithms are used in decision-making. Yet in our design, preferences for or against algorithms as such should merely produce level effects on delegation rates. Our between-subject treatment comparison holds the algorithm, the task, and the decision context fixed and varies only whether punishment is possible, so that such level effects difference out.

Responsibility perceptions of decisions involving algorithms. Within this body of work, a smaller subset focuses on algorithm use preferences in settings whose consequences fall partly or wholly on someone other than the decision-maker, sometimes called the *moral domain* (Gogoll and Uhl, 2018). Much of this work concludes that aversion to algorithms is amplified once third-party welfare is at stake. Gogoll and Uhl (2018) document this in an incentivized calculation task with performance uncertainty, Bigman and Gray (2018) in vignettes across high-stakes moral domains (driving, legal, medical, military), and Jauernig et al. (2022) identify moral discretion as the human capacity that people are most reluctant to hand over in such contexts.⁴

Once decisions affect others, the responsibility-shifting motive established for delegation to humans may become relevant also for algorithmic delegation, and with it the question of how observers apportion blame when an algorithm is involved in the decision. A first strand of evidence, predominantly vignette-based and unincentivized, finds leniency toward the human decision-maker. Physicians who follow standard-of-care AI recommendations are judged less harshly, and face lower legal exposure, than physicians who reject them, among lay and expert evaluators alike (Tobia et al., 2021; Tacconelli et al., 2026). Algorithmic discrimination in hiring triggers less moral outrage than identical human discrimination (Bigman et al., 2023). In autonomous driving, blame tracks the allocation of control. The same harm is judged more leniently when an autonomous vehicle rather than a human driver causes it (Gill, 2020), whereas under shared control drivers receive more blame than their vehicles, even for crashes they could not reasonably avoid (Awad et al., 2020; Beckers et al., 2022). The link between control and blame extends beyond driving, with humans who merely monitor an algorithm blamed less than humans who decide alone (Shank et al., 2019).

A second strand uses incentivized experiments. Much of it finds that algorithmic intermediaries can serve as moral cover. Extending the canonical dictator game of Bartling and Fischbacher (2012), Chevrier and Teixeira (2024) find that delegation to an algorithm lowers expected

³Interest in the comparison between human and statistical judgment is long-standing. See Meehl (1954) and Dawes (1979) for the classic demonstrations that simple statistical rules outperform clinical judgment and are resisted nonetheless.

⁴For recent surveys of this area, see Bonnefon et al. (2024) and, with a legal-policy focus, Sunstein and Gaffe (2025).

punishment relative to choosing the same inequalitarian allocation directly. Compared with delegation to a human intermediary, recipients punish less often following algorithmic delegation but impose more punishment on the delegator when they do. In a similar setting, [Tontrup and Sprigman \(2025\)](#) let dictators hand the execution of a transfer they chose to a large language model. Transfers fall when this option is available, and dictators as well as outside observers attribute part of the responsibility to the machine. [Köbis et al. \(2025\)](#) find that delegation to algorithms increases dishonest behavior. In a consequential donation choice with real moral stakes, [Hüholt and Szech \(2026\)](#) find that subjects delegate more often when an AI rather than a human is available and that delegation is associated with lower reported responsibility. However, not all studies find evidence of such cover. [Ismagilova and Ploner \(2025\)](#) study an investment task in which agents either choose investments on behalf of principals themselves or delegate the choice to an algorithm. Principals’ decisions to reward or punish agents depend on the realized return, but this outcome dependence does not differ significantly between delegated and non-delegated decisions.

The study closest to ours is [Feier et al. \(2022\)](#). In their design, a subject determines the payoff of another participant either through her own performance in a Raven-style logic task or, depending on the treatment, through delegation to another human or to an algorithm. The affected participant can reward or punish the subject for the decision and its result. They find that after good outcomes delegated and self-made decisions are rewarded similarly in both treatments. After bad outcomes, decisions delegated to the algorithm, but not decisions delegated to another human, are rewarded more generously than self-made ones. Delegation rates do not differ significantly across treatments.

Our design departs from theirs in three respects. First, their algorithm is calibrated at the session level, so the delegation decision signals the subject’s beliefs about her own ability relative to the session performance distribution. As a result, punishment may partly sanction perceived overconfidence, outweighing any downstream punishment as a measure of responsibility shifting.⁵ We calibrate the algorithm at the individual level, which removes the relative-performance signal from the delegation decision by construction. Second, in their design the same subject acts as both delegator and evaluator, which can import motives of self-justification or hypocrisy aversion into the punishment decision.⁶ We separate the two roles across disjoint subject pools. Third, their design holds the punishment possibility fixed across treatments, so it cannot identify whether the prospect of punishment motivates delegation in the first place. Any difference in delegation rates in their design may thus be due to general preferences for algorithm use. We vary the punishment possibility directly. Our experiment therefore provides, to our knowledge, the first direct test of how the prospect of punishment by an affected party influences delegation to an algorithm.

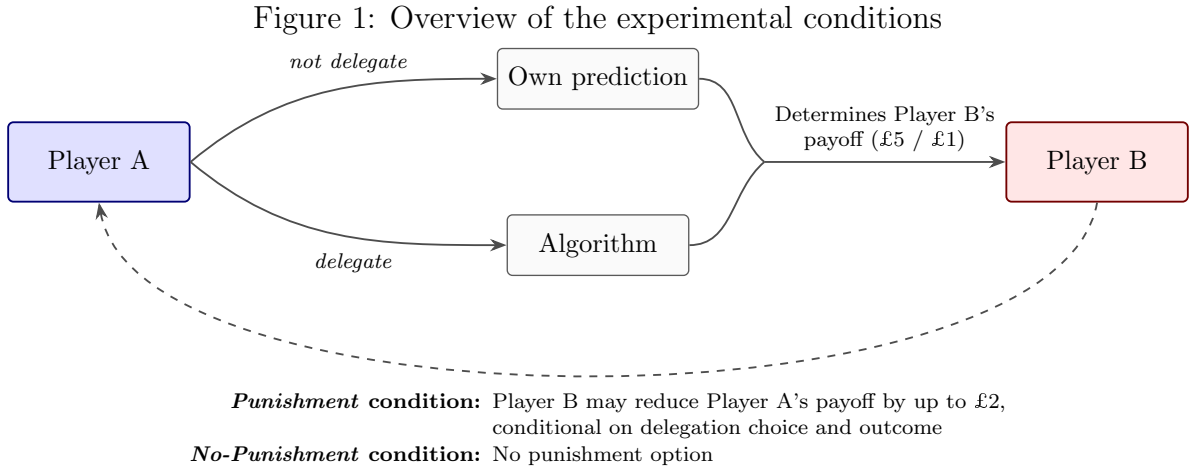
⁵Their own analysis shows that the propensity to delegate is driven primarily by beliefs about task performance, not by the type of delegate.

⁶In their data, for example, evaluators who themselves delegated evaluate delegated decisions more favorably than self-made ones after good outcomes, whereas non-delegating evaluators favor self-made decisions ($r = -0.34$, $p = 0.004$, machine treatment; own computations from their published data).

3 Experimental Design

We ran an online experiment via the British platform Prolific (<https://www.prolific.com>) with UK participants. The experiment was programmed in oTree (Chen et al., 2016) and eligibility followed Prolific’s standard quality screens, a minimum 95% prior approval rate, a fluent-English filter, and a desktop-only restriction (no mobile or tablet) that kept the screen layout consistent across subjects.

The design combines a between-subject manipulation of whether costly punishment is possible with a within-subject elicitation of conditional punishment behavior. Within each condition, participants assumed one of two roles. Player A decides whether to make a payoff-relevant prediction for Player B herself, or to delegate the prediction to an algorithm calibrated to match her own measured performance. Player B is the matched recipient whose payoff is fully determined by the outcome of the prediction, whether Player A makes it herself or delegates it to the algorithm. In the *Punishment* condition Player B may subsequently punish Player A, conditional on both the delegation decision and the realized outcome, while in the *No-Punishment* condition no punishment is possible. Figure 1 summarizes the basic setup.



Procedure

Player A

After providing informed consent, Player A completed ten incentivized rounds of a visual prediction task. Specifically, in each round, she predicted a person’s weight from a facial image and the person’s height. She could enter predictions via a slider or type them directly in kilograms, pounds, or stones-and-pounds.⁷ No feedback was provided between rounds. At the end of the experiment, one of the ten rounds was randomly selected for payment. Player A earned £4 if her prediction in that round was within 10 lbs of the true weight and £2 otherwise. These

⁷All individuals depicted in the images had explicitly consented to the use of their photographs for research purposes.

initial rounds served three purposes simultaneously: First, they established a stake from which any later punishment could be deducted. Second, they familiarized Player A with the task. Third, they generated the performance data needed to calibrate the algorithm.

The choice of a visual prediction task is not incidental. Algorithm preferences may depend sensitively on the algorithm’s perceived relative competence (Qin et al., 2025; Logg et al., 2019). Our design neutralizes this concern by calibrating the algorithm to match each subject’s individual performance. The visual nature of the task lends credibility to this equal-performance claim, as people have everyday familiarity with judging others’ build and appearance, and algorithmic error is plausible.⁸ In contrast, subjects may be skeptical that an algorithm could ever fail in purely mathematical tasks (Gogoll and Uhl, 2018; Feier et al., 2022). Finally, we chose a prediction task involving genuine uncertainty rather than a setting with an obvious self-serving motive (as in classic dictator-game variations). This reflects the nature of many real-world applications of algorithms, in which outcomes are inherently uncertain, the decision-maker has no direct material interest in the affected party’s outcome, and performance is evaluated ex post.

After the ten rounds, we randomly paired Player A with another participant, Player B, who had not completed the prediction task himself. We informed Player A that her final prediction would not affect her own payoff but would almost entirely determine Player B’s. A prediction within 10 lbs of the true weight would give Player B a high payoff (£5), and any less accurate prediction a low payoff (£1).⁹ Player A then learned that she could either make the prediction herself or delegate it to an algorithm. She was also told that, based on her first ten predictions, the algorithm would have performed equally well.

We calibrated the algorithm to mimic Player A’s own performance over the first ten rounds. Concretely, if Player A delegated the final prediction to the algorithm, the algorithm was equivalent to drawing a Bernoulli outcome with success probability equal to Player A’s empirical accuracy rate over her first ten rounds. For example, a subject who scored 3 out of 10 in the initial rounds thus delegates to an algorithm that produces the high payoff for Player B with 30% probability.

Fixing the algorithm’s success probability to each subject’s own accuracy makes the equal-performance claim hold subject by subject, not merely on average across a pool of participants. This matters because beliefs about relative performance are otherwise likely a dominant driver of delegation.¹⁰ For example, under the session-level calibration of Feier et al. (2022), the delegation choice is largely driven by each subject’s sense of her ability relative to the algorithm. Matching the algorithm to each subject neutralizes this motive, and making this performance

⁸Predicting body weight from an image has been used in earlier experiments (Logg et al., 2019).

⁹Since the final prediction determines only Player B’s payoff, Player A has no financial interest in Player B receiving the low payoff. As long as a better outcome for Player B weakly reduces Player A’s expected punishment in the *Punishment* condition, the incentives of the two players can be considered aligned.

¹⁰This motive would not inherently bias the treatment-effect comparison, since it should be balanced across the *Punishment* and *No-Punishment* conditions, but it might otherwise dominate the delegation decision and may obscure the responsibility-shifting motive of interest.

parity common knowledge should limit Player B from reading delegation as a signal of Player A’s believed relative competence. This makes any difference in punishment easier to interpret as a response to delegation itself rather than to perceived overconfidence.

In the *Punishment* condition, we further informed Player A that Player B could impose (costly) punishment of up to £2, and could condition punishment on both her delegation decision and the realized outcome. In the *No-Punishment* condition, Player B had no such punishment option. The two delegation options (own decision / algorithm) appeared in random order across subjects. Following prior work, we attach no monetary cost to delegation (Bartling and Fischbacher, 2012; Coffman, 2011; Feier et al., 2022).¹¹ If Player A chose to make the prediction herself, she entered her final prediction on the next screen. If she delegated instead, this screen was skipped.

We then asked Player A to report her beliefs about Player B’s punishment in each of the four scenarios resulting from the combination of delegation decision (yes/no) and prediction outcome (high/low payoff for Player B). We elicited beliefs on the same scale as actual punishment for Player B (£0–£2 in £0.10 increments) and presented the four scenarios in block-wise random order. We chose not to incentivize belief elicitation for three reasons. First, only two of the four scenarios remain payoff-relevant after the delegation decision, making the others purely hypothetical. Second, an incentivized mechanism could induce hedging. Third, an incentive-compatible scoring rule across four scenarios would have substantially increased task complexity. Because we are nonetheless interested in comparing punishment beliefs to actual punishment, we included an attention check on the belief-elicitation screen. In the *No-Punishment* condition, where no actual punishment was possible, we elicited beliefs about hypothetical punishment to keep the experimental flow identical across conditions.

We then asked Player A to assess her own performance in the first ten rounds by assigning a probability mass (summing to 100%) to each possible score from 0 to 10.¹² We elicited this belief after the delegation decision, and after Player A had been told that the algorithm would perform equally well. While learning about relative algorithm performance may lead her to revise her beliefs about her own performance, it is precisely this post-update belief that is the analytically relevant input for any subsequent behavior. We also asked Player A to estimate the distribution of scores in the population, assigning to each score from 0 to 10 the share of participants she expected to have achieved it. Combined with her belief about her own score, this locates where she places herself in the distribution and yields a measure of perceived ability relative to other participants. One score was drawn at random at the end of the experiment, and Player A earned £0.30 if her estimate for it fell within five percentage points of the realized

¹¹This allows the delegation decision in the *No-Punishment* condition to more closely reflect Player A’s intrinsic preference over delegation. Moreover, if delegation carried a positive cost, any responsibility-shifting motive would first have to outweigh that cost.

¹²The elicitation was incentivized. Player A earned an additional £0.30 with the probability she had assigned to her actual score. While this linear scoring rule is not proper as its payoff-maximizing report places all mass on the single most likely score, irrespective of risk attitude, in practice most subjects spread probability mass across several scores.

population share.

Finally, we elicited risk preferences using an incentivized multiple price list in the style of [Holt and Laury \(2002\)](#) and administered a short questionnaire on socio-demographic characteristics and technology affinity, before presenting Player A with her preliminary payoff.

Player B

After providing informed consent, Player B learned that he had been randomly matched with a Player A, and introduced to the task Player A had completed. He learned that his own payoff would be determined by the outcome of Player A’s final prediction – £5 if it fell within 10 lbs of the true weight and £1 otherwise – while Player A’s payoff depended solely on her performance in the initial ten rounds. Crucially, we also told Player B that Player A had the option to delegate the final prediction to an algorithm that performs as well as Player A had performed in the earlier rounds, and that Player A was aware of this, making it common knowledge.

In the *Punishment* condition, Player B could choose to reduce Player A’s payoff by up to £2 in £0.10 increments. We focus on punishment rather than reward because the responsibility-attribution literature has predominantly operationalized blame through punishment ([Bartling and Fischbacher, 2012](#); [Coffman, 2011](#); [Steffel et al., 2016](#)). Imposing strictly positive punishment required Player B to forgo £0.10 of his own earnings, ensuring that punishment was costly to the punisher. We elicited punishment decisions for all four combinations of delegation (yes/no) and outcome (high/low payoff for Player B), implementing only the decision matching the realized combination. Because the punishment decision is the central outcome variable for Player B, we included an attention check on the punishment-elicitation screen. In the *No-Punishment* condition, where actual punishment was not possible, we elicited hypothetical punishment decisions on the same scale. Player B’s perceptions of task difficulty, his risk preferences, and a short questionnaire followed, before we presented him with his final payoff.

After Player B’s decisions were recorded, we calculated Player A’s final payoff and paid all participants on the day of participation.

Hypotheses and identification

Our design tests two main hypotheses, both grounded in the literature on responsibility avoidance through delegation to human intermediaries ([Bartling and Fischbacher, 2012](#); [Coffman, 2011](#); [Oexl and Grossman, 2013](#); [Hamman et al., 2010](#); [Hill, 2015](#); [Steffel et al., 2016](#)) and on responsibility attributions to algorithmic decision-makers ([Feier et al., 2022](#); [Gogoll and Uhl, 2018](#)). These strands of the literature mostly focus on what [Gogoll and Uhl \(2018\)](#) have coined the *moral* domain. Although the underlying prediction task is not itself moral, Player A’s choice imposes monetary consequences on a matched recipient, and a poor prediction causes real monetary harm. In this respect, our setting shares the defining feature of the settings this literature

studies. The responsibility-shifting account predicts that delegation protects Player A from punishment following an adverse outcome. If Player A anticipates this protection and does not expect it to be offset by greater punishment for delegation following a favorable outcome, the possibility of punishment should make delegation more attractive. Accordingly, we formulate Hypothesis 1.

Hypothesis 1 (Delegation). *The introduction of potential punishment increases the likelihood that Player A delegates the final prediction to the algorithm.*

Hypothesis 1 is grounded in empirical evidence that responsibility avoidance shapes delegation decisions involving human intermediaries. Bartling and Fischbacher (2012) also show that subjects are more likely to delegate morally consequential decisions even to a die roll, an early form of delegation to a non-intentional decision mechanism, suggesting that the motive of responsibility avoidance persists when the intermediary lacks intentionality. Whether the same logic extends to algorithmic delegates, to decisions under genuine uncertainty, and to settings in which the outcome falls entirely on the recipient is less explored. Feier et al. (2022) report that delegation rates are similar for human and algorithmic intermediaries, suggesting that the kind of intermediary may matter less than the mere availability of one. Their design, however, does not compare delegation rates across punishment regimes, leaving open the question Hypothesis 1 addresses. Hypothesis 1 is identified by the between-subject comparison of Player A’s delegation rate across the *Punishment* and *No-Punishment* conditions, which exploits the punishment manipulation as the only between-subject difference and holds fixed every other feature of the decision environment.

The delegation response predicted by Hypothesis 1 rests on Player A anticipating protection from punishment following an adverse outcome. This gives rise to our second hypothesis.

Hypothesis 2 (Punishment). *Conditional on a low-payoff outcome, delegation reduces Player B’s punishment of Player A.*

Evidence for Hypothesis 2 comes from Bartling and Fischbacher (2012), who find that delegation to a randomization device allows decision-makers to avoid some punishment following an unfair outcome, although less so than delegation to another human. Similarly, Feier et al. (2022) find that decision-makers receive more favorable evaluations following bad outcomes when they delegated to an algorithm than when they relied on their own performance.

Existing evidence also offers some support for the absence of an offsetting punishment premium for delegation following favorable outcomes. Feier et al. (2022) find no significant difference between evaluations of delegated and own decisions following good outcomes, and in the dictator game settings with human intermediaries, Bartling and Fischbacher (2012) and Coffman (2011) find no significant increase in punishment for intermediation when the resulting allocation is fair. Moreover, although Coffman (2011) show in a separate experiment that intermediation can reduce the reward assigned to prosocial behavior, receiving less credit need not translate into punishment when there is no adverse consequence to sanction.

Hypothesis 2 is identified by the within-subject comparison between delegated and own predictions following a low-payoff outcome. The corresponding comparison following a high-payoff outcome provides evidence on whether delegation itself attracts additional punishment, independent of the outcome. Both comparisons isolate the effect of delegation on punishment while holding fixed both the realized consequences for Player B and all between-subject heterogeneity in Player B’s punishment proclivity.

4 Results

Data collection took place between February and June 2023 through the British online platform Prolific, with participation restricted to UK residents. A total of 322 subjects were recruited (161 assigned to Player A, 161 to Player B).¹³ On average, Player A spent 12 minutes completing the experiment and earned £3.62, while Player B spent approximately 8 minutes and earned £2.40. The treatment groups are balanced on the background characteristics reported in Appendix A.1, Table 3.

In the following, we first test the two main hypotheses derived from our preregistered research questions (Results 1-2). The subsequent analyses serve to understand the mechanism behind these results (Results 3-5). They are exploratory in nature, though several of them were outlined as secondary analyses in the preregistration.¹⁴

Player A – Delegation behavior

We begin by examining the delegation behavior of Player A. In the *Punishment* condition, where Player B can punish, 42.5% of Player As delegated the final prediction to the algorithm. In the *No-Punishment* condition, where punishment is impossible, 60.8% delegated. The introduction of potential punishment thus significantly *reduces* delegation by 18.3 percentage points, opposing the direction posited by Hypothesis 1.¹⁵ Figure 2 illustrates the difference. Delegation

¹³We preregistered the collection of data from 320 participants (80 subjects per role per treatment). Due to attrition and subsequent re-matching we recruited an extra pair in the *No-Punishment* condition. For the Player A analyses, we exclude two subjects in the *No-Punishment* condition because they entered a prediction of 0 lbs. in at least eight of the ten initial rounds.

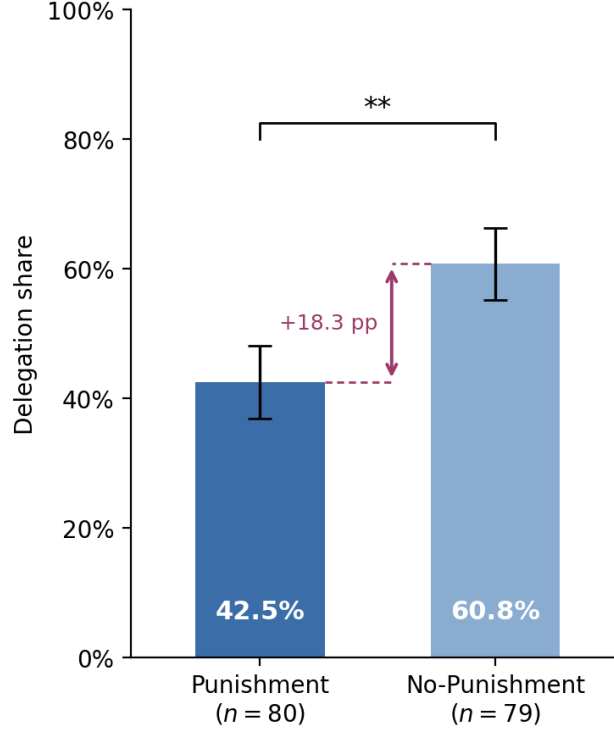
¹⁴Unless stated otherwise, all tests are two-sided. We describe results as significant if $p < 0.05$ and as weakly significant if $0.05 \leq p < 0.10$. Because the delegation decision is a binary outcome compared between conditions, delegation shares are tested using Pearson Chi-squared tests, with Fisher’s exact tests yielding the same conclusions throughout. Because punishment is elicited within-subject through the strategy method, punishment comparisons use paired *t*-tests, with Wilcoxon signed-rank tests as non-parametric counterparts. Regressions report robust (HC1) standard errors. The preregistration and any deviations from the preregistered analyses are described in Appendix C.

¹⁵A within-subject corroborator points in the same direction. Player As in the *No-Punishment* condition were asked, after their actual delegation decision, the hypothetical question of whether they would have delegated had punishment been possible. Among the 79 respondents, 12 would have stopped delegating under hypothetical punishment (25% of delegators) while only 5 would have started (16.1% of non-delegators), constituting weakly significant switching behavior towards making the prediction oneself with punishment possible (within-subject McNemar’s test $p = 0.090$), and yielding a hypothetical delegation rate of 51.9%.

shares around 50% are consistent with subjects, on average, finding the claim of equal relative performance between themselves and the algorithm credible. Their behavior reflects neither systematic preference for one’s own prediction nor systematic deference to the algorithm.

Result 1. *The possibility of punishment significantly reduces the likelihood that Player A delegates the final prediction to the algorithm.*

Figure 2: Delegation shares by condition



Note: Bars show the share of Player As who delegated the final prediction to the algorithm in each condition. Whiskers are ± 1 standard error of the mean. The treatment difference is significant at the 5% level (Pearson Chi-squared test, $p = 0.021$).

We next examine the robustness of the observed treatment difference in delegation shares. Table 1 presents Probit estimates of the delegation decision. Column (1) reports the unconditional treatment effect. Column (2) adds socio-demographic and other background characteristics. Columns (3) and (4) add Player A’s actual performance (number of correct predictions in the first ten rounds) and her belief about her own performance, respectively. Column (5) restricts the sample to subjects who passed the attention check.¹⁶ The treatment effect is robust across all five specifications. Average marginal effects indicate that introducing the punishment possibility reduces the probability of delegation by 18.3 to 21.8 percentage points. Actual performance is negatively and significantly associated with delegation, whereas the relationship with perceived performance is weaker. We return to this pattern when discussing the mechanisms.

¹⁶While the attention check was included on another screen, it may still correlate with who took the delegation decision seriously.

Table 1: Determinants of the delegation decision

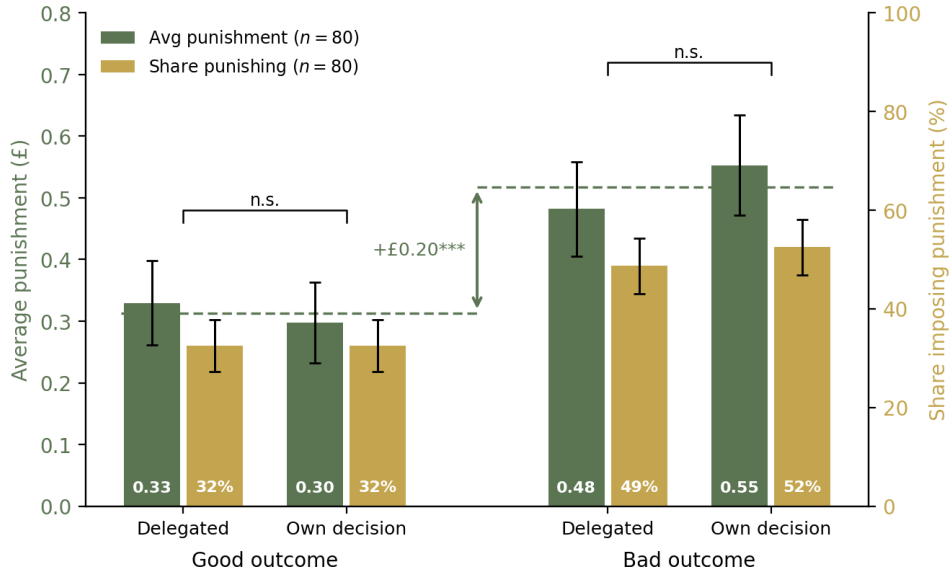
	Dependent variable: <i>Delegation</i>				
	<i>Full sample</i>				<i>Passers</i>
	(1)	(2)	(3)	(4)	(5)
Punishment indicator	−0.462** (0.201)	−0.473** (0.206)	−0.537** (0.209)	−0.489** (0.207)	−0.589** (0.235)
Task performance			−0.160** (0.081)		−0.203** (0.089)
Perceived performance				−0.075 (0.057)	
Age		−0.003 (0.009)	−0.002 (0.009)	−0.002 (0.009)	0.002 (0.011)
Female		0.141 (0.228)	0.194 (0.230)	0.181 (0.231)	0.372 (0.257)
Socio-economic status		−0.044 (0.071)	−0.042 (0.073)	−0.036 (0.072)	−0.086 (0.081)
Went to uni		0.032 (0.228)	0.062 (0.226)	0.020 (0.227)	0.118 (0.248)
Technology score		−0.065 (0.093)	−0.036 (0.096)	−0.056 (0.093)	0.024 (0.109)
Leadership position		0.415* (0.226)	0.444* (0.228)	0.411* (0.226)	0.278 (0.256)
Constant	0.273* (0.143)	0.533 (0.557)	0.908 (0.593)	0.801 (0.606)	0.851 (0.673)
AME of Punishment (pp)	−18.3** ($p = 0.019$)	−18.3** ($p = 0.019$)	−20.3*** ($p = 0.008$)	−18.7** ($p = 0.016$)	−21.8*** ($p = 0.009$)
N	159	157	157	157	128
Pseudo R^2	0.024	0.042	0.060	0.051	0.077

Note: Coefficients from a Probit regression of the delegation decision on the Punishment indicator and a set of controls. Robust (HC1) standard errors in parentheses. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Due to missing socio-demographic data, two subjects are dropped in Columns (2)–(5). The *AME of Punishment* reports the average change in the predicted probability of delegation from introducing the punishment possibility, in percentage points.

Player B – Punishment behavior

We now turn to Player B’s punishment behavior in the *Punishment* condition, in which punishment was payoff-relevant. Punishment is substantial in all four (delegation, outcome) scenarios of the strategy method, with average punishment between £0.30 and £0.55 (Figure 3, left axis). Punishment responds strongly to the outcome. Player Bs punish bad outcomes significantly more than good outcomes, by £0.20 on average, indicating that measured punishment carries real informational content rather than being random. The substantial punishment after good outcomes need not reflect noise either. Player Bs can hold views about how the final prediction should have been made, independently of the realized outcome, and can punish accordingly.¹⁷

Figure 3: Player B punishment by delegation and outcome scenario



Note: For each (delegation, outcome) scenario, the green bars report the average punishment chosen by Player B (left *y*-axis, in pounds) and the gold bars the share of Player Bs imposing non-zero punishment (right *y*-axis, in percent), in the *Punishment* condition. Whiskers are ± 1 standard error of the mean. The brackets mark the within-outcome comparisons of average punishment between delegated and Player A’s own predictions. The dashed lines mark the mean of average punishment within each outcome group. The subsample of attention-check passers is shown in Appendix Figure 6. Hypothetical punishment in the *No-Punishment* condition is reported in Appendix Figure 8.

Holding the outcome fixed, Player B punishes delegated and own decisions no differently. After a good outcome, average punishment is £0.33 for delegated and £0.30 for own decisions (paired *t*-test, $p = 0.53$; Wilcoxon signed-rank $p = 0.27$). After a bad outcome, £0.48 versus £0.55. The punishment gap for bad outcomes thus runs in the direction predicted by Hypothesis 2 but is small and statistically insignificant (£0.07; paired *t*-test $p = 0.13$; Wilcoxon signed-rank $p = 0.54$).

Result 2. *Conditional on a low-payoff outcome, Player B does not punish delegated decisions significantly less than Player A’s own decisions.*

¹⁷For example, Coffman (2011) likewise documents non-trivial punishment in settings where the decision-maker’s action could not have been improved upon.

Regression analysis in Appendix A.2 Table 4 confirms this pattern. Punishment loads overwhelmingly on the outcome, while the effect of delegation is essentially zero.¹⁸ The same pattern holds on the extensive margin (Appendix A.3 Table 5). For each outcome, the share of Player Bs imposing any punishment is indistinguishable between delegated and own decisions (Figure 3, right axis). Punishment is also far from universal. Nearly half of Player Bs (45.0%) impose no punishment in any of the four scenarios. Within each scenario, average punishment therefore reflects a punishing minority rather than light, diffuse sanctioning.

In sum, Player B punishes delegated decisions neither less harshly nor less frequently than Player A’s own decisions. In our setting, Player A can therefore not escape punishment by delegating the final prediction to the algorithm, but she also pays no punishment premium for doing so. Punishment responds almost entirely to the realized outcome.¹⁹ This result also bears on Result 1 as any treatment difference in delegation frequencies driven by punishment avoidance would be inconsistent with Player B’s observed punishment behavior.

Mechanisms

Result 1 does not merely fail to support Hypothesis 1, but the effect is statistically significant in the opposite direction. The possibility of punishment discourages, rather than encourages, delegation to the algorithm. Thus, the prospect of punishment alters Player A’s delegation decision, even though we detect no punishment penalty for delegating. In the remainder of this section, we examine the data more closely and weigh candidate explanations with the goal of reconciling the two main results.

Departures from canonical settings. Hypothesis 1 drew on evidence from settings that differ from ours in ways that plausibly matter. In the modified dictator games of [Bartling and Fischbacher \(2012\)](#); [Coffman \(2011\)](#); [Hamman et al. \(2010\)](#); [Oexl and Grossman \(2013\)](#), the decision-maker chooses between a fair and an unfair allocation. Her intent is therefore transparent, the chosen allocation is implemented with certainty, and the unfair option benefits her directly at the recipient’s expense. Punishment in these settings sanctions a deliberate, self-serving act, and delegation is attractive precisely because it blurs the link between the principal and that act. Our prediction task departs from these settings. The outcome is uncertain, so a bad outcome may reflect bad luck rather than bad intent, and neither option affects Player A’s own monetary payoff, leaving no self-serving choice to police. With little intent

¹⁸In the regression, because punishment for the four scenarios is elicited within-subject, any stable characteristic of Player B is differenced out and cannot, by construction, confound the within-subject comparison. Thus, the control variables merely capture the between-subject association with Player B’s average punishment level. We find no striking relationship between Player B’s average punishment level and either their socio-demographic characteristics or their beliefs about Player A’s absolute performance.

¹⁹Hypothetical punishment choices elicited from Player Bs in the *No-Punishment* condition corroborate this conclusion (Appendix Figure 8). If anything, these Player Bs state that they would punish a delegated decision more than a decision made directly by Player A after a bad outcome (£0.53 versus £0.43; paired *t*-test, $p = 0.069$; Wilcoxon signed-rank $p = 0.11$).

worth sanctioning, any intention-based punishment motive may lose its bite, and with it does the value to Player A of placing an algorithm between herself and the outcome. We nevertheless chose this task deliberately since prediction under genuine uncertainty, by a decision-maker with no direct material stake in the affected party’s outcome, is the environment in which real-world algorithmic delegation typically operates. These departures, however, may explain the absence of the predicted effect, not its reversal. A reversal requires a motive that weighs against delegation when punishment is possible.

Anticipated differential punishment. The most natural candidate explanation is anticipation of differential punishment. Player A may expect a delegated decision to be punished more harshly than an own decision, and delegate less in response. Such an expectation would, for example, be consistent with previous evidence that observers view the delegation of morally consequential decisions to machines relatively critically (e.g., Gogoll and Uhl, 2018; Bigman and Gray, 2018; Jauernig et al., 2022; Bonnefon et al., 2024).

After the delegation decision but before learning the outcome, we elicited Player A’s beliefs about Player B’s punishment in each of the four (delegation, outcome) scenarios. Anticipated punishment can affect behavior only where punishment is possible. We therefore focus the analysis of punishment beliefs on the *Punishment* condition. Two patterns emerge (Figure 4). First, Player As, on average, substantially overestimate punishment in every scenario. Second, Player As expect essentially the same punishment for delegated and own decisions, conditional on the outcome, correctly anticipating the pattern in Player B’s actual punishment behavior (Result 2). Thus, elicited beliefs provide no support for a punishment anticipation mechanism for Result 1.²⁰

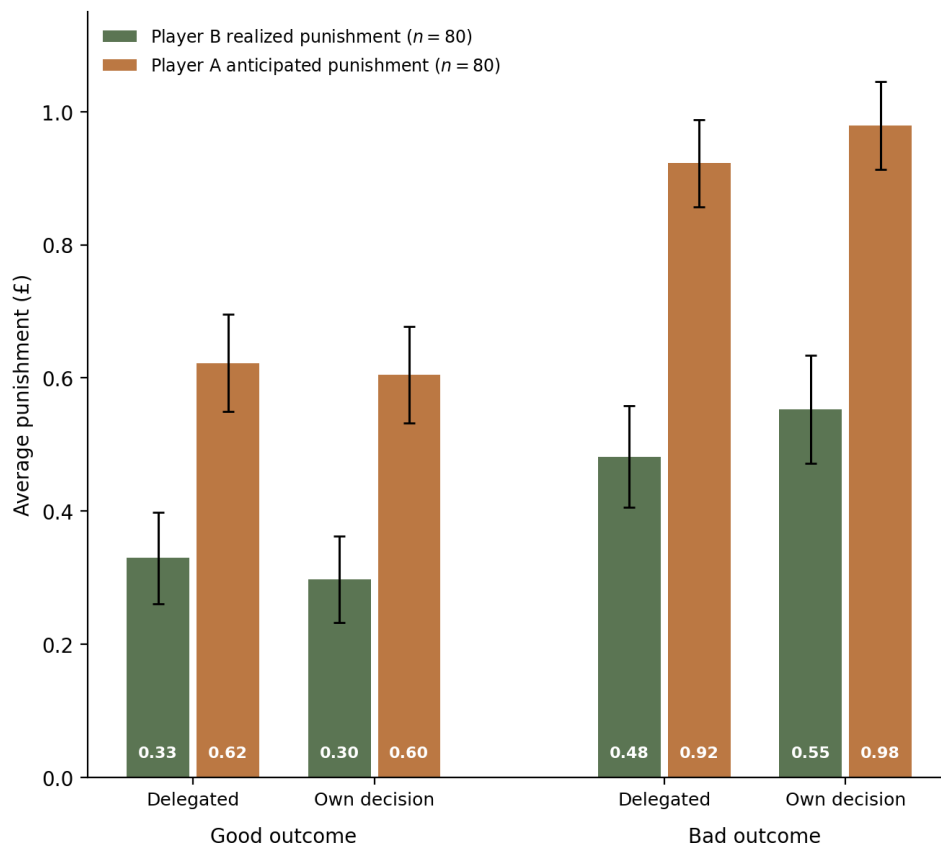
Player As, however, are not unanimous in the punishment penalty they expect. We therefore ask whether these individual differences predict delegation. Table 2 reports Probit regressions of the delegation decision on Player A’s belief difference in expected punishment (*no delegation* – *delegation*), so that higher values correspond to a larger expected punishment penalty for deciding herself. All specifications are restricted to the *Punishment* condition. Columns (1)–(3) use the simple average of the belief differences for good and bad outcomes, Column (4) replaces the simple average with a weighted average, using Player A’s elicited probability of producing the high payoff as the weight, and Column (5) enters the two differences separately.

Expected differential punishment does not predict delegation in the full sample (Columns (1)–(2)). Among subjects who passed the attention check on the belief-elicitation screen, the relationship sharpens. A £0.10 larger expected punishment penalty for deciding herself is associated with a 4.7 percentage-point higher probability of delegation (Column (3)), and with 5.5 percentage points under the weighted specification (Column (4)).²¹

²⁰The same holds for the fully hypothetical beliefs elicited in the *No-Punishment* condition, which closely track those in the *Punishment* condition (Appendix Figure 8). The treatment manipulation itself therefore appears not to have changed what subjects expected punishment to be.

²¹In the decomposition of Column (5), the good-outcome belief difference carries the relationship (2.7 per-

Figure 4: Realized vs. anticipated punishment, by scenario



Note: Player B's realized punishment (green bars) and Player A's anticipated punishment (terracotta bars), in pounds, across the four (delegation, outcome) scenarios in the *Punishment* condition (full sample). Whiskers denote ± 1 standard error of the mean. The subsample of attention-check passers is shown in Appendix Figure 7.

Table 2: Delegation and beliefs about punishment

	Dependent variable: <i>Delegation</i>				
	<i>Full Punishment</i>		<i>Punishment, passers</i>		
	(1)	(2)	(3)	(4)	(5)
Belief difference, average ($\bar{\Delta}$)	0.346 (0.407)	0.398 (0.448)	1.661** (0.700)		
Belief difference, weighted by $\Pr(\text{good})$ (Δ^w)				1.992** (0.794)	
Belief difference, bad outcome (Δ^{bad})					0.590 (0.424)
Belief difference, good outcome (Δ^{good})					0.961** (0.412)
Task performance		-0.208* (0.111)	-0.362*** (0.136)	-0.342** (0.142)	-0.381*** (0.138)
Age		0.023* (0.012)	0.039** (0.016)	0.040** (0.016)	0.039** (0.016)
Female		0.152 (0.350)	0.950** (0.427)	1.026** (0.406)	0.845* (0.444)
Socio-economic status		-0.049 (0.111)	-0.006 (0.131)	-0.055 (0.133)	0.001 (0.126)
Went to uni		0.071 (0.363)	-0.278 (0.412)	-0.169 (0.408)	-0.275 (0.409)
Technology score		-0.061 (0.138)	0.048 (0.170)	0.072 (0.173)	0.033 (0.167)
Leadership position		0.302 (0.312)	-0.301 (0.408)	-0.279 (0.408)	-0.245 (0.411)
Constant	-0.198 (0.142)	-0.345 (0.826)	-1.228 (1.155)	-1.281 (1.147)	-1.115 (1.165)
AME of belief measure (pp per £0.10)	1.3 ($p = 0.388$)	1.4 ($p = 0.370$)	4.7** ($p = 0.011$)	5.5*** ($p = 0.006$)	
AME: good-outcome difference (pp per £0.10)					2.7** ($p = 0.010$)
AME: bad-outcome difference (pp per £0.10)					1.7 ($p = 0.165$)
Controls	×	✓	✓	✓	✓
Passers only	×	×	✓	✓	✓
N	80	78	60	60	60
Pseudo R^2	0.007	0.098	0.248	0.263	0.252

Note: Probit estimates with robust (HC1) standard errors in parentheses. Sample restricted to the *Punishment* condition throughout. Belief differences are within-subject (*no delegation*) – (*delegation*) in expected punishment, so positive coefficients indicate that subjects who expect to be punished more harshly for not delegating (relative to delegating) are more likely to delegate. Columns (1)–(3) use the simple average of the good-outcome and the bad-outcome belief difference. Column (4) instead weights the good-outcome difference by Player A’s elicited probability of producing the high payoff and the bad-outcome difference by the complementary probability. Column (5) enters the two differences separately. The *AME* rows report the average marginal effect of the belief measure entered in the respective column on the probability of delegation, in percentage points per £0.10 increase in the corresponding belief difference.

Taken together, the belief evidence does not support anticipated differential punishment as the mechanism behind Result 1. For anticipation of punishment to generate an 18.3 percentage-point delegation gap between conditions, Player As would need to expect systematically harsher punishment for delegated decisions. The measured beliefs show no such expectation, and such an expectation would not have been correct, as Player B’s realized punishment exhibits no delegation penalty either (Result 2). What the belief evidence does show is heterogeneity. Within the *Punishment* condition, Player As whose beliefs embody a larger penalty for deciding themselves are more likely to delegate. Anticipated punishment is thus a dimension along which subjects differ, but it provides no aggregate push toward less delegation under punishment.²² We nonetheless read the belief evidence with caution. First, hypotheticality may introduce noise as only two of the four scenarios could still materialize after Player A’s delegation decision in the *Punishment* condition.²³ Second, the elicitation of beliefs was unincentivized and took place after the delegation decision, so reports may reflect post-hoc rationalization rather than the beliefs that drove the choice.

Result 3. *On average, Player A expects no differential punishment for delegation, mirroring Player B’s actual punishment behavior.*

Sample composition. Result 1 is not an artifact of sample composition either. Treatment randomization produced balance across socio-demographic and other background characteristics (Appendix A.1, Table 3). Initial task performance, however, is weakly higher in the *No-Punishment* condition. Player As averaged 2.76 versus 3.15 correct predictions out of ten in the *Punishment* and *No-Punishment* conditions, respectively (t -test, $p = 0.059$; Figure 5). However, this difference works against Result 1. As we have seen, lower-performing Player As are more likely to delegate, so that the lower average performance in the *Punishment* condition should raise delegation there and therefore attenuate, rather than generate, the observed treatment difference. Player As in both conditions report similar perceived performance (4.65 versus 4.89, respectively).²⁴ Finally, the treatment effect is robust to controlling for actual and perceived performance (Table 1, Columns (3)–(5)).

Performance and delegation. In the Probit specifications, task performance is negatively and significantly associated with delegation (Table 1, Columns (3) and (5)). Descriptively,

centage points per £0.10, $p = 0.010$), while the bad-outcome difference is positive but noisier (1.7 percentage points, $p = 0.165$). The two differences are nearly orthogonal in this sample, so we read their coefficients as additive partial slopes rather than competitors for the same variance. Under expected-utility reasoning, however, only their probability-weighted sum should enter the decision rule. We therefore treat Column (5) merely as a descriptive decomposition.

²²A mean expected penalty of zero need not rule out an effect on delegation if Player As react more strongly when they expect a penalty for delegating than when they expect one for deciding themselves. We find no such asymmetry.

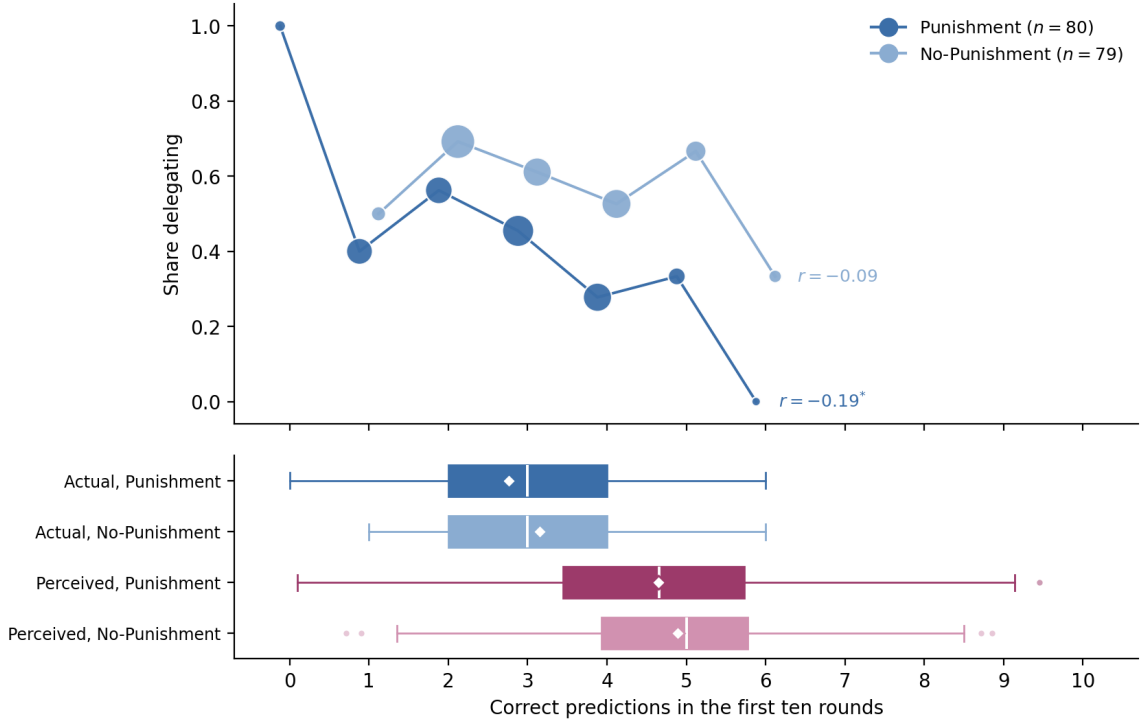
²³In the *No-Punishment* condition all four scenarios were purely hypothetical.

²⁴Perceived performance is measured by the weighted-average belief about own performance. This overconfidence may partly reflect the information that the algorithm performs equally well.

this relationship is stronger in the *Punishment* condition than in the *No-Punishment* condition (Figure 5), although the difference between the two relationships is not statistically significant. For perceived performance, a directionally similar but weaker relationship emerges in both conditions. Thus, actual performance separates delegators from non-delegators and because the relationship is slightly stronger in the *Punishment* condition the delegation gap between conditions widens with performance. Player As who delegated in the *Punishment* condition scored on average 2.47 out of 10 in the first ten rounds, compared with 3.06 for delegators in the *No-Punishment* condition (t -test, $p = 0.039$). It is precisely the better performers who, particularly when punishment is on the table, keep the decision for themselves.

Result 4. *Better-performing Player As are less likely to delegate.*

Figure 5: Delegation, performance, and confidence by condition



Note: The top panel shows the delegation share of Player A at each score, by treatment. Dot size is proportional to the number of subjects at that score. The bottom panel shows the distributions of actual and perceived (the weighted-average belief about own performance) scores, by condition. Boxes span the interquartile range, the white line marks the median, and the white diamond the mean.

Our individual-level calibration is designed to remove the natural explanation for this relationship, namely that worse performers delegate because they believe the algorithm outperforms them. For example, in [Feier et al. \(2022\)](#), the algorithm’s performance distribution is matched to the distribution of human performance at the population level. A high-performing subject who recognizes her position in that distribution therefore rationally expects to outperform the algorithm, and a low-performing subject rationally expects the reverse. Delegation that declines with performance is then exactly what expected-payoff maximization predicts. In our experiment, by contrast, performance is equalized between the algorithm and each individual subject, and this equality of performance is common knowledge. The negative relationship

between performance and delegation therefore points to something other than beliefs about relative ability.

Two mechanisms may account for the negative association between performance and delegation. The first is a preference over *process ownership*. Player As with a higher probability of producing the high payoff may prefer to bring it about themselves rather than have the algorithm do so, making delegation less attractive as performance rises. A symmetric version of this preference, under which Player As also wish to avoid personally producing a low payoff, yields the same negative relationship. Relatedly, subjects' preferences may depend on *outcome uncertainty*. Dietvorst and Bharti (2020) document that the preference for algorithms declines as outcome uncertainty rises. In our setting, uncertainty is arguably greatest when the probability of success is near one half. Because most subjects score below 5 out of 10, better performance moves that probability toward this point. A reluctance to delegate more uncertain decisions could therefore likewise make better-performing Player As less likely to delegate.

Both readings describe motives that should operate similarly in both conditions. They can therefore account for the performance-delegation relationship, but not on their own for the treatment difference in delegation. Indeed, evaluated against this negative relationship, the slightly lower average performance in the *Punishment* condition would, if anything, imply more delegation. It is nevertheless conceivable that the prospect of punishment makes these concerns more salient. For example, even with equal expected punishment, being punished for a decision one gave away may feel different from being punished for a decision one made oneself. Or relinquishing control over an uncertain outcome may trouble Player A only once that outcome carries indirect consequences for Player A. The descriptively stronger performance–delegation relationship under punishment is consistent with such a treatment dependence in the salience of these motives. However, the treatment-specific relationships do not differ statistically. Moreover, greater salience alone does not determine the aggregate effect. For example, a stronger symmetric preference over process ownership would tend to encourage delegation among the predominantly low-performing Player As, whereas an asymmetric desire to produce the good outcome oneself or a heightened reluctance to delegate uncertain decisions could discourage it. We therefore regard these motives merely as possible explanations for the performance-delegation relationship, but not as established explanations for the treatment difference.

Effort to outperform the algorithm. Lastly, the prospect of punishment could motivate Player A to exert additional effort when making the final prediction for Player B. Although the algorithm's chance of success is fixed by Player A's past performance, the final prediction itself has yet to be made. Player A may thus believe that, by trying especially hard, she can outperform the algorithm and thereby give Player B a better chance of the high payoff. To the extent that she expects milder punishment after a good outcome, the prospect of punishment would then push her toward making the prediction herself. This story yields a testable prediction. Relative to the first ten rounds, non-delegators should improve their performance more in the *Punishment* condition than in the *No-Punishment* condition. Similar improvement in

both conditions would instead indicate a punishment-independent motive, for instance altruism toward Player B, which cannot explain Result 1.

The data do not support this story. Measuring improvement as the difference between a subject’s average absolute prediction error over the first ten rounds and her absolute error on the final prediction, non-delegators in the *No-Punishment* condition improve, whereas non-delegators in the *Punishment* condition do not.²⁵ Because all Player As faced the same images in the first ten rounds and on the final prediction, this contrast cannot reflect variation in image difficulty. Realized outcomes tell the same story. 48.4% of non-delegators in the *No-Punishment* condition earn the high payoff for Player B, compared with 21.7% in the *Punishment* condition, even though the two groups’ performance over the first ten rounds is statistically indistinguishable (3.29 vs. 2.98 correct predictions; *t*-test, $p = 0.316$).²⁶ If anything, performance on the final prediction is higher where punishment is unavailable, the opposite of what a punishment-motivated effort account predicts.²⁷ Effort itself remains unobserved, however, and absolute prediction errors, realized outcomes as well as time spent on the decision screen are only imperfect measures of it, so we cannot rule the channel out conclusively.²⁸

Result 5. *The prospect of punishment does not induce additional effort.*

The signaling value of not delegating. Although the prospect of punishment does not appear to induce additional effort (Result 5), the delegation choice itself may communicate effort. By keeping the decision, Player A may want to convey to Player B that she takes the task seriously and intends to try hard on his behalf. Delegating, by contrast, could be perceived as shirking. The signal lies in the choice itself rather than in any effort that follows, so Result 5 does not speak against this account. Result 3, however, seems to. The signal has monetary value only if Player A expects delegation, as apparent shirking, to draw harsher punishment, whereas in the *No-Punishment* condition, Player A could shirk without monetary consequences. Result 3 indicates that Player As in the *Punishment* condition, on average, hold no such expectation, which leaves little monetary gain from sending the signal.²⁹

Two considerations keep this channel open. First, Player A may care about sending this signal,

²⁵Non-delegators in the *No-Punishment* condition reduce their prediction error by 8.11 lbs on average, compared with only 0.51 lbs in the *Punishment* condition (Welch *t*-test, $p = 0.027$; Mann–Whitney test, $p = 0.014$). Their average initial absolute errors are virtually identical (22.69 versus 22.47 lbs, respectively; Welch *t*-test, $p = 0.857$)

²⁶Balance among non-delegators across conditions coexists with the selection pattern rather than contradicting it. Non-delegators in the *Punishment* condition are drawn from a slightly weaker pool but are more strongly positively selected within it, so the two forces partly offset, whereas among delegators they compound.

²⁷Employing screen time as a measure for effort points in the same direction. In the *Punishment* condition Player As spend less time on the final prediction screen than in the *No-Punishment* condition (medians 16 vs. 23 seconds). The same holds within subject, as non-delegators in both conditions spend more time on the final prediction than on an average initial round, but the increase is smaller under punishment (+5.0 vs. +8.0 seconds).

²⁸For instance, if the prospect of punishment raised effort while the accompanying pressure depressed performance, additional effort would be present, yet masked.

²⁹At the individual level, this channel is consistent with the belief heterogeneity documented in Result 3, though a mean expected penalty of zero implies no aggregate push toward less delegation.

beyond any punishment attached to it. For this concern to explain the treatment difference, however, it would again have to be activated by the prospect of punishment itself. In our environment, where no self-serving option exists and outcomes are noisy, the delegation choice is the only deliberate act on which Player B can condition punishment, and thus the natural focus of any judgment he forms. Second, as discussed earlier, our punishment belief measure was unincentivized and elicited only after the delegation decision, so it may capture the beliefs behind that decision imperfectly and understate the penalty Player A anticipated for delegating. Either way, a concern about being perceived as shirking would weigh against delegation precisely when punishment is possible.

Taking stock. The treatment difference in delegation shares (Result 1) is robust, but its mechanism remains unresolved. It is not a composition artifact, since the conditions are balanced in background characteristics, while the weak imbalance in task performance works against the treatment effect. Belief evidence suggests that anticipated differential punishment cannot explain the reversal either (Result 3). On average, Player A expects no punishment penalty for delegation, mirroring Player B’s realized punishment (Result 2), although Player As who expect a larger penalty for deciding themselves are more likely to delegate. Nor do we find evidence that punishment-induced effort can explain the reversal. Even though actual effort remains unobserved, non-delegators spend less time on the prediction for Player B and improve less under punishment, not more (Result 5). What the data do show is selection. It is the better-performing Player As who retain the decision (Result 4). Preferences over process ownership or outcome uncertainty are consistent with this selection but operate in both conditions alike. The signaling value of not delegating requires no actual additional effort (Result 5), however, to explain the treatment effect it rests on being activated by the prospect of punishment itself.

5 Conclusion

As algorithms take over decisions that affect other people, who should answer for the results has become the subject of a lively legal, ethical, and policy debate. This paper examines the behavioral side of this debate. We ask whether the prospect of being held responsible shapes the decision to put an algorithm in charge, and whether putting an algorithm in charge shapes the responsibility assignment of those affected by the outcome. We take both questions to a preregistered online experiment in which a decision-maker either makes a payoff-relevant prediction for a recipient herself or delegates it to an algorithm calibrated, subject by subject, to match her own performance. Between subjects, we vary whether the recipient can reduce the decision-maker’s earnings. We elicit punishment for every combination of delegation choice and prediction outcome, so we observe directly whether delegation shields the delegator from punishment, conditional on the outcome. Because the two conditions differ only in whether

punishment is possible, the comparison of delegation rates identifies the effect of the prospect of punishment on the decision to delegate.

Both answers depart from the canonical account of delegation as a means of responsibility shifting. The prospect of punishment does not push decision-makers toward the algorithm but away from it, a reversal of the prediction imported from settings with human intermediaries (e.g., [Bartling and Fischbacher, 2012](#)). Delegation, in turn, offers no protection against punishment. Recipients punish delegated and Player A's own predictions alike, conditioning their sanctions almost entirely on the realized outcome rather than on how the prediction was made. The punishment result offers an explanation for why the canonical prediction fails, yet not for its reversal.

We examine several explanations to reconcile these findings. Beliefs show no punishment premium attached to delegation. Prediction outcomes and screen times do not indicate punishment-induced additional effort, whereby the decision-maker would retain the decision in the hope of outperforming the algorithm. Merely signaling to exert additional effort by retaining the decision would require no measurable effort at all, but move delegation only if the prospect of punishment gave the signal value. Belief evidence speaks against such a monetary value, leaving only a non-monetary value in which the prospect of punishment makes the recipient's judgment salient in itself. However, both measures are imperfect. The belief elicitation was unincentivized and came only after the delegation decision, so it may capture post-hoc rationalization rather than the beliefs that drove the choice, while actual effort is unobserved, with prediction outcomes and screen times serving only as proxies. The measures may therefore simply fail to detect a decision-maker who fears that delegating will be read as shirking and draw harsher punishment, or who retains the decision to try to outperform the algorithm, a motive that would at least be consistent with recipients punishing bad outcomes more heavily. These limits leave the mechanism narrowed rather than settled.

Nevertheless, we contribute by providing the first direct test of whether the prospect of punishment motivates delegation to an algorithm, and show that in an environment with inherent task uncertainty and no self-serving choice to police, accountability disciplines algorithm use rather than fueling it. We also show that delegation cannot shift blame when the algorithm performs no better or worse than the decision-maker, with recipients conditioning their sanctions on the outcome rather than on how the prediction was made. More broadly, our findings add a determinant of algorithm use to those documented in the aversion literature, one that operates not through the algorithm or the task but through what delegating may convey about the person who chooses it.

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Appendix

A Additional Tables

A.1 Condition balance — Player A

Table 3: Condition balance — Player A

Variable	<i>Punishment</i>	<i>No-Punishment</i>	<i>p-value</i>
Age	39.026	37.886	0.557
Female	0.487	0.506	0.936
Socio-economic status	5.179	5.329	0.542
Went to uni	0.575	0.557	0.945
Technology score	2.087	2.266	0.366
Leadership position	0.362	0.329	0.783

Note: Mean values by condition for Player A, with p -values from two-sided t -tests for continuous variables and Chi-squared tests for binary variables. *Socio-economic status* is self-reported on a 1–10 scale. *Went to uni* indicates a university degree. *Technology score* is constructed from weekly device usage, programming skills, cryptocurrency knowledge, technology use at work, and similar items. *Leadership position* indicates self-reported management or supervisory experience.

Complementary to the treatment comparisons of individual observable characteristics in Table 3, we can test their joint significance by regressing the treatment indicator on all characteristics in a linear probability model. The observables are jointly uninformative about treatment ($F(6, 150) = 0.37$, $p = 0.90$; $R^2 = 0.01$; $N = 157$). A probit specification yields the same conclusion (likelihood-ratio $\chi^2(6) = 2.30$, $p = 0.89$).

We do not separately report a balance table for Player B. Player B’s punishment behavior is identified within-subject across the four (delegation, outcome) scenarios of the strategy method, so balance across conditions is not strictly required for the punishment analysis.

A.2 Regression analysis of punishment

Table 4 complements the paired comparisons in the main text with a stacked OLS regression of Player B’s chosen punishment on delegation, the outcome, and their interaction. The coefficient on *Delegated* captures the punishment difference between delegated and own predictions after a good outcome, while the interaction captures how that difference changes after a bad outcome, and thus any insulation from punishment after bad outcomes. The delegation coefficient is close to zero in both samples. The interaction is negative and weakly significant -0.103 ($p = 0.077$) in the full sample and -0.078 ($p = 0.138$) among attention-check passers. Like the paired comparisons, the regression indicates that punishment responds primarily to the realized outcome and provides no strong evidence that delegation insulates Player A from punishment.

Table 4: Determinants of Player B's chosen punishment

	Dependent variable: <i>Punishment</i> (£)	
	<i>Full sample</i>	<i>Passers</i>
	(1)	(2)
Delegated	0.032 (0.052)	0.001 (0.054)
Bad outcome	0.256*** (0.071)	0.260*** (0.080)
Delegated $\times$ Bad outcome	-0.103* (0.058)	-0.078 (0.052)
Player B belief about Player A perf.	-0.020 (0.032)	0.001 (0.037)
Age	0.002 (0.005)	-0.001 (0.006)
Female	-0.100 (0.128)	-0.064 (0.140)
Socio-economic status	0.083* (0.047)	0.072 (0.049)
Went to uni	-0.196 (0.140)	-0.098 (0.142)
Technology score	0.073 (0.053)	0.030 (0.059)
Leadership position	0.083 (0.142)	0.061 (0.164)
Constant	-0.077 (0.328)	0.020 (0.374)
Observations	320	268
Subjects	80	67
Adj. R^2	0.077	0.030

Note: OLS regression of Player B's chosen punishment levels. The sample is stacked over the four (delegation, outcome) scenarios of the strategy method (4 observations per Player B). Column (1) uses the full Punishment-condition sample. Column (2) restricts to Player Bs who passed the attention check on the punishment-elicitation screen. Standard errors clustered at the Player B level.

A.3 Extensive margin of punishment

Table 5 re-estimates the punishment regression on the *extensive* margin, replacing the punishment amount with an indicator for whether Player B imposed any punishment in a given scenario (Probit). The conclusion matches the intensive-margin result in the body. A bad outcome sharply raises the probability of punishment, whereas delegation leaves it essentially unchanged. Player A is therefore no less *likely* to be punished when she delegates, just as she is punished no less *harshly*.

Table 5: Extensive margin of punishment — Probit

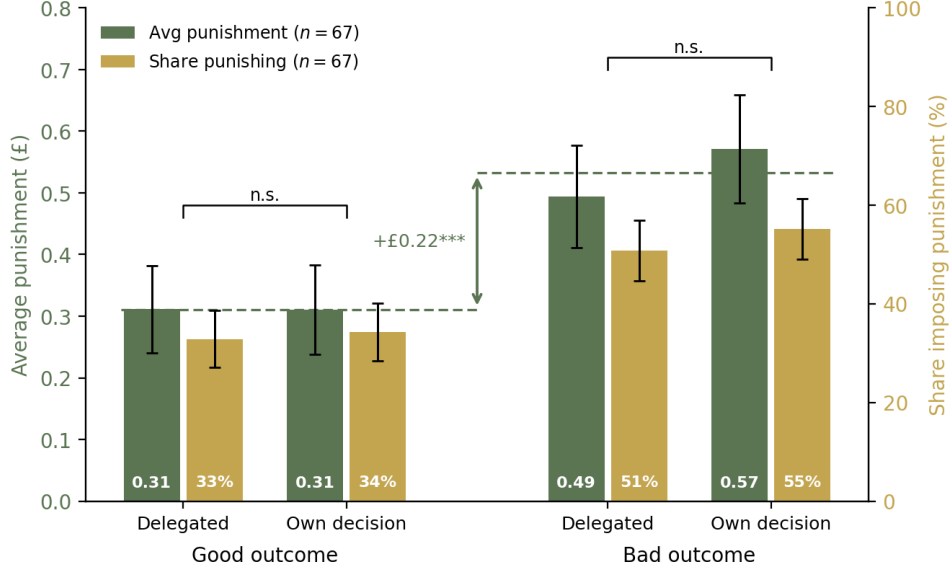
	Dependent variable: <i>Any punishment</i>	
	<i>Full sample</i>	<i>Passers</i>
	(1)	(2)
Delegated	0.002 (0.077)	−0.042 (0.078)
Bad outcome	0.556*** (0.139)	0.563*** (0.153)
Delegated × Bad outcome	−0.099 (0.095)	−0.072 (0.103)
Player B belief about Player A perf.	−0.083 (0.075)	−0.030 (0.082)
Age	−0.006 (0.013)	−0.013 (0.015)
Female	0.115 (0.272)	0.230 (0.291)
Socio-economic status	0.160 (0.126)	0.140 (0.129)
Went to uni	−0.827*** (0.320)	−0.706** (0.349)
Technology score	0.062 (0.118)	−0.023 (0.131)
Leadership position	0.062 (0.293)	−0.071 (0.321)
Constant	−0.382 (0.807)	−0.182 (0.864)
AME of Delegated (pp)	−1.8 ($p = 0.462$)	−2.9 ($p = 0.285$)
AME of Bad outcome (pp)	18.1*** ($p < 0.001$)	19.4*** ($p < 0.001$)
Observations	320	268
Subjects	80	67
Pseudo R^2	0.096	0.081

Note: Probit regression of an indicator for non-zero punishment. The sample is stacked over the four (delegation, outcome) cells of the strategy method (4 observations per Player B). Column (1) uses the full Punishment-condition sample; Column (2) restricts to attention-check passers. Standard errors clustered at the Player B level. The *AME* rows report discrete average marginal effects on the probability of imposing any punishment, in percentage points. AME p-values use a 400-replication subject-cluster bootstrap. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

B Additional figures

B.1 Punishment among attention-check passers

Figure 6: Player B punishment by delegation and outcome scenario, attention-check passers

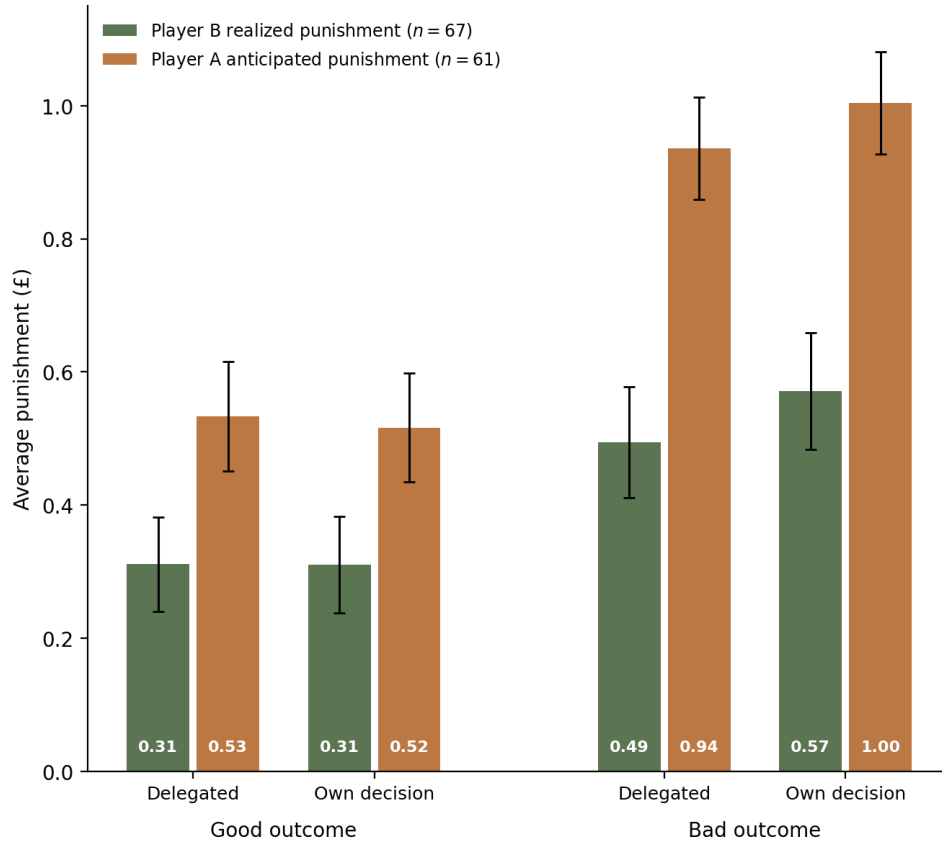


Note: As in Figure 3, but restricted to Player Bs who passed the attention check on the punishment-elicitation screen ($n = 67$). The four (delegation, outcome) scenarios of the strategy method are grouped by outcome. For each scenario, the green bars report the average punishment (left y -axis, in pounds, whiskers are ± 1 standard error of the mean) and the gold bars the share of Player Bs imposing non-zero punishment (right y -axis, in percent, whiskers are ± 1 standard error of the share). The brackets mark the within-outcome comparisons of average punishment between delegated and Player A's own predictions. The dashed lines mark the mean of average punishment within each outcome group, and the arrow reports their difference.

B.2 Hypothetical punishment

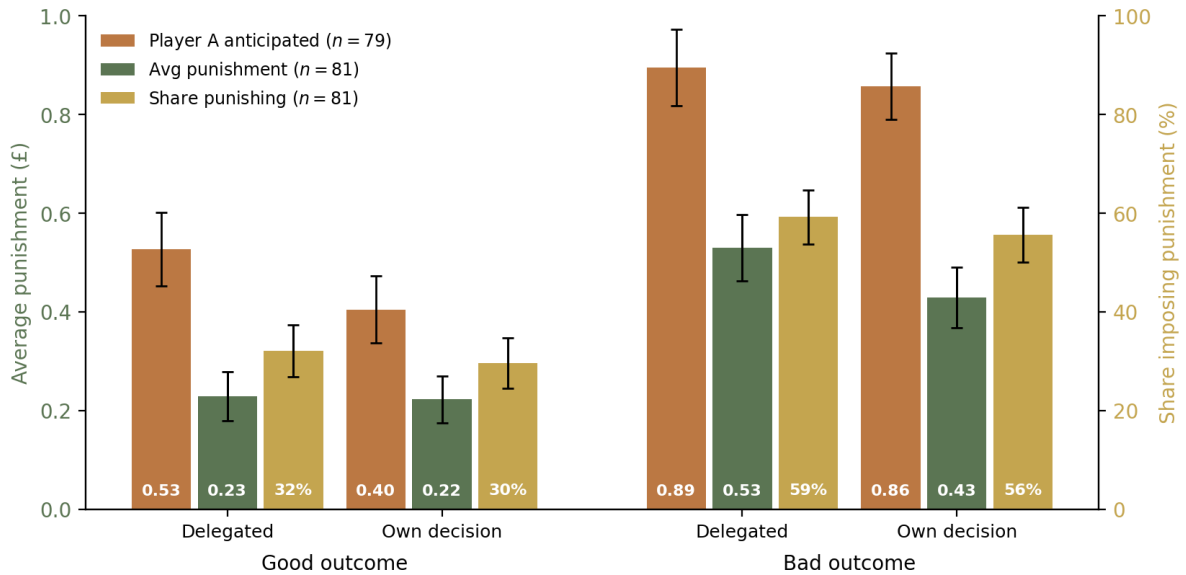
Hypothetical punishment in the *No-Punishment* condition follows the same outcome-driven pattern as actual punishment (Figure 8). After good outcomes, stated punishment is nearly identical for delegated and non-delegated decisions (£0.23 versus £0.22; paired t -test, $p = 0.91$; Wilcoxon signed-rank $p = 0.78$). After bad outcomes, the comparison reverses relative to the direction predicted by Hypothesis 2. Player Bs state that they would punish a delegated decision more than a decision made directly by Player A (£0.53 versus £0.43; paired t -test, $p = 0.069$; Wilcoxon signed-rank $p = 0.11$). Even in stated terms, delegation thus offers Player A no insulation from punishment. Player A's anticipated punishment, shown alongside, overstates hypothetical punishment in every scenario, mirroring the overestimation of actual punishment in the *Punishment* condition (Figure 4).

Figure 7: Realized vs. anticipated punishment, attention-check passers



Note: As in Figure 4, but restricted to attention-check passers (Player A $n = 61$, Player B $n = 67$).

Figure 8: Hypothetical punishment by delegation and outcome scenario (*No-Punishment* condition)



Note: As in Figure 3, but for the *No-Punishment* condition, in which Player Bs answered the same punishment questions but could not impose punishment in practice. The four (delegation, outcome) scenarios are grouped by outcome. For each scenario, the green bars report the average hypothetical punishment (left y -axis, in pounds; whiskers are ± 1 standard error of the mean) and the gold bars report the share of Player Bs choosing non-zero punishment (right y -axis, in percent; whiskers are ± 1 standard error of the share). In addition, the terracotta bars report Player A's hypothetical anticipated punishment in the *No-Punishment* condition. Values are printed at the base of each bar.

C Preregistration

The study was preregistered on AsPredicted under the title “Algorithms and Responsibility” (#133980) on May 30, 2023, after the collection of a pilot sample and before the collection of the main sample.

The analyses in the paper deviate from the preregistration in three respects. First, we recruited 322 participants instead of the planned 320. Attrition during the experiment required re-matching and produced one additional pair in the *No-Punishment* condition. Throughout the analysis, we exclude two Player As, because they submitted a prediction of 0 lbs. in at least eight of the ten initial rounds. Second, we compare delegation rates across conditions using a Pearson chi-squared test and Fisher’s exact test rather than the preregistered *t*-test. The delegation decision is binary, so these tests are the appropriate choice, and the conclusion is unchanged under either procedure. Third, the preregistration specifies Mann-Whitney U- and Kolmogorov-Smirnov tests for the punishment comparisons. Both are tests for independent samples, whereas punishment is elicited within-subject via the strategy method. We therefore report the Wilcoxon signed-rank test as the within-subject non-parametric counterpart alongside the parametric tests. Beyond these deviations, the preregistration left open whether participants who failed the attention checks would be excluded. We present the analysis for both samples.

Algorithms and Responsibility (#133980)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

Pre-registered on:
2023/05/30 16:27 (PT)

1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

We test whether:

RQ 1. punishment for a decision can be avoided if it is delegated to an algorithm (conditional on performance)

RQ 2. a punishment motive enters the delegation decision.

3) Describe the key dependent variable(s) specifying how they will be measured.

The experiment features a delegator and an evaluator. The delegator has to make a decision that is payoff-relevant only to the evaluator. The delegator may decide herself or delegate the decision to an algorithm, which is calibrated to match the delegators performance. Depending on the treatment, the evaluator may punish the delegator conditional on the outcome of the decision and the delegation decision (strategy method).

For RQ 1, we will look at evaluator punishment behavior and compare the amount and frequency of punishment between a delegated and non-delegated decision conditional on the outcome.

For RQ 2, we will look at the delegation frequency between the treatments (punishment possible vs punishment not possible).

4) How many and which conditions will participants be assigned to?

There are two conditions (punishment possible/no punishment possible) and two roles in each condition (delegator/evaluator).

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

I will run parametric (t-test) and non-parametric (MWU, Kolmogorov-Smirnov) tests between punishment (amount and frequency) for delegated and non-delegated outcomes, conditional on the outcome. Similarly, I will run linear regressions with punishment behavior as the dependent variable and explore whether expected performance or any socio-demographic characteristics are predictive of differential punishment.

For delegators, I will again, run parametric (t-test) tests on delegation choices between treatments. Moreover, I will employ Probit regressions with the dependent variable being the delegation decision and a set of explanatory variables including treatment, beliefs about performance and socio-demographic characteristics.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

I will exclude people that access the experiment via mobile device. Moreover, I may exclude participants that failed the attention checks.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

I will recruit subjects via Prolific. I plan to recruit 320 participants, 160 per treatment, 80 per role per treatment.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

I already ran a pilot with 25 observations in the punishment treatment. I plan to include the data in the analysis.

Secondary analysis will include an investigation on how performance and beliefs about performance impact results. Moreover, I will look at the delegator's beliefs about expected punishment and their consistency with the delegation decision.

At this point, I already want to state that I may run another treatment in which the delegator may either delegate to another participant or an algorithm.